ODA-Bench UAV Obstacle Avoidance

Real-log offline evaluation for planning, risk and policy learning

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Real logs

ODA trajectories, sensor CSV, obstacle metadata.

Offline benchmark

Evaluate planners and lightweight learned policies before expensive field tests.

Abstract: ODA-Bench turns real logs into a benchmark

Problem

- UAV obstacle avoidance is risky to test directly because crashes are expensive.
- Simulation is useful, but sim-to-real gap makes offline real logs valuable.
- ODA provides real UAV logs, but not yet a standardized offline benchmark protocol.

Proposal

- Convert ODA logs into geometry, labels, tasks and baselines.
- Evaluate planners, BC policies and risk predictors offline.
- Validate candidate ranking with a lightweight simulator harness.

●● **Insight.** Contribution nằm ở benchmark construction + evaluator + tasks + metrics + baselines + validation, không phải tạo raw ODA dataset mới.

The gap is between raw datasets and usable safety evaluation

Real logs

Trajectory and sensor evidence from real UAV flights, but no common task protocol.

Simulation

Closed-loop testing is controllable, but realism depends on sensor and dynamics fidelity.

Field tests

Most realistic, but slow, expensive and unsafe for early-stage candidates.

Raw ODA logs → ODA-Bench protocol → Ranked candidates

Insight. ODA-Bench là lớp chuẩn hóa ở giữa: biến log thật thành metric có thể so sánh và dùng để lọc ứng viên trước simulator/field test.

Our contributions are benchmark components, not a new raw dataset

1. We convert ODA real UAV logs into a standardized offline benchmark.
2. We define collision, clearance, violation, latency-aware and learning metrics.
3. We provide planner, imitation learning and risk prediction tracks with lightweight baselines.
4. We implement a simulation validation harness for ranking candidates before field testing.

Current evidence

300 ODA trial CSV subset, 14,700 expert samples, 4 baseline families, 3 main plots.

Scope boundary

No claim of field safety or PX4 closed-loop safety. PyBullet backend is implemented; execution is pending on a compatible machine.

Related work leaves a real-log benchmark gap

Existing directions

- UAV obstacle avoidance datasets capture real sensor/trajectory evidence.
- Simulation benchmarks test closed-loop controllers under controlled settings.
- Learning-based avoidance evaluates policies, often with synthetic or simulator-heavy data.
- Offline RL benchmarks define logged transitions and cost/reward protocols.

What is missing

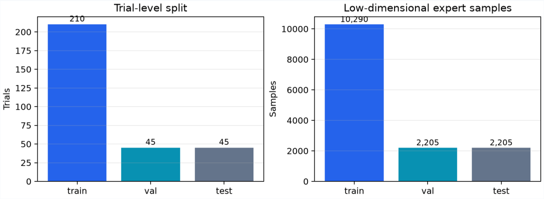
- A trial-level split and evaluator on real UAV logs.
- Shared safety labels for collision, violation and clearance.
- Comparable tracks for planners, BC and risk models.
- A lightweight bridge from offline ranking to simulator validation.

Insight. Unlike prior workflows, ODA-Bench uses real UAV logs and supports both offline evaluation and lightweight offline policy training.

ODA-Bench construction: from CSV logs to benchmark cases

Raw input used

- 300 local ODA trials downloaded from Hugging Face.
- 300 `optitrack.csv`, 300 `imu.csv`, 300 `radar.csv`.
- Trial metadata gives obstacle position, light condition and video flags.



Geometry proxy

$$\mathbf{p}_t = (x_t, z_t), \qquad \mathbf{o}_j = (x_j, z_j)$$

$$c_t = \min_j \|\mathbf{p}_t - \mathbf{o}_j\|_2 - r_{\text{obs}}$$

Split is by trial ID, not by random timestep, to reduce leakage across train/val/test.

Evaluation labels separate contact from near-miss risk

Clearance over a trajectory

$$c_{\text{min}} = \min_t c_t$$

Safety labels

$$\text{collision} = \mathbb{1}[c_{\text{min}} \leq 0]$$

$$\text{violation} = \mathbb{1}[0 < c_{\text{min}} < r_{\text{safe}}]$$

Constants

Symbol	Value	Meaning
r_{obs}	0.20 m	contact radius
r_{safe}	0.50 m	safety margin
$r_{\text{obs}} + r_{\text{safe}}$	0.70 m	violation boundary

Latency proxy

$$d_{\text{delay}} = \|\mathbf{v}\| T_{\text{delay}}$$

$$T_{\text{delay}} = T_{\text{sensor}} + T_{\text{map}} + T_{\text{planner}} + T_{\text{cmd}}$$

Insight. Collision is contact. Violation is a near-miss inside the safety ring, which is useful for ranking before actual crashes happen.

The benchmark defines four downstream tracks

Track	Input	Output	Primary metrics
Offline planner	start, goal, obstacles	path/trajectory	collision, violation, clearance, length, runtime
Filtered BC	expert trajectories	local waypoint action	action MSE, rollout safety, inference latency
Risk prediction	state/sensor features	future-risk score	PR-AUC, recall, false negative, calibration
Offline RL	logged transitions	learned policy	return, cost, collision, sim validation

Main claim tracks

Planner, BC and risk are fully run on the 300-trial CSV subset.

Stretch track

Offline RL is kept as future work until TD3+BC/IQL/Decision Transformer baselines are validated.

Baselines are intentionally lightweight and reproducible

Main baselines

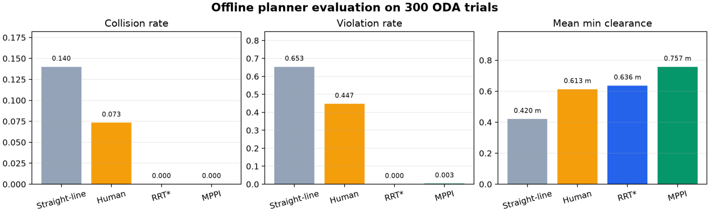
- Planner: RRT* and MPPI.
- Behavior cloning: Plain BC-MPPI and Filtered BC-MPPI.
- Risk: TTC/clearance threshold and Small MLP.
- Validation: ODA-like obstacle course, 20 seeds $\times$ 4 speeds.

Kept out of main story

- RRT, A*, straight-line and human trajectory: useful appendix references.
- Heavy offline RL: not a main claim until simulation results are stable.
- Real flight safety: explicitly out of scope.

●● **Insight.** The benchmark is designed to run on MacBook-class hardware first, then validate finalists in a simulator.

RQ1: ODA-Bench separates unsafe baselines from safe planners

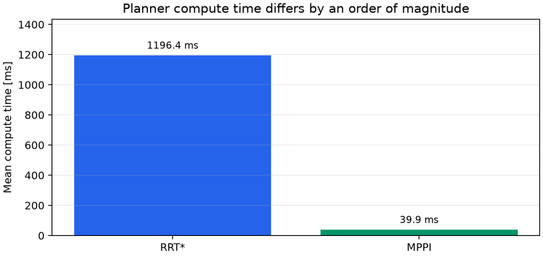


Main readout

- Straight-line has high collision and violation.
- RRT* reaches zero collision and zero violation on the 300-trial geometry benchmark.
- MPPI reaches zero collision and the highest clearance among main planners.

Human/OptiTrack is a real flight trajectory reference, not an optimal planner. Main comparison should focus on RRT* vs MPPI.

Planner safety and compute trade off differently

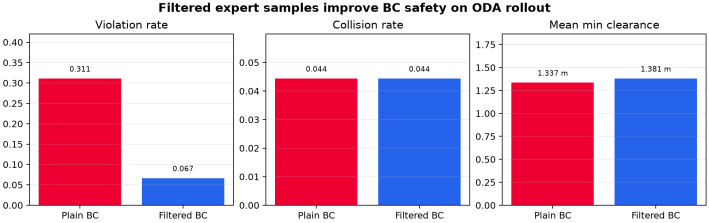


RRT* vs MPPI on 300 ODA trials

- RRT*: collision 0.0000, violation 0.0000, clearance 0.6360 m.
- MPPI: collision 0.0000, violation 0.0033, clearance 0.7574 m.
- Mean compute time: RRT* 1196.4 ms vs MPPI 39.9 ms.

Insight. Offline safety alone is not enough for online UAV use. Compute and sensor delay determine whether a safe path arrives in time.

RQ2: filtering expert samples improves BC safety



300-trial split

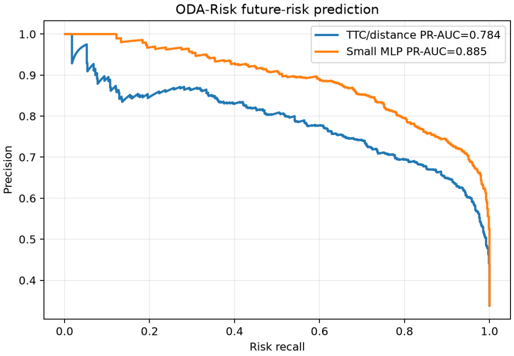
- Train: 210 trials, 10,290 samples.
- Val: 45 trials, 2,205 samples.
- Test: 45 trials, 2,205 samples.

Result

- Violation drops from 0.3111 to 0.0667.
- Clearance increases from 1.3365 m to 1.3809 m.

Collision rate is tied at 0.0444; filtered BC improves near-miss behavior more clearly than contact rate on this split.

RQ3: ODA risk labels support useful future-risk prediction



Test set

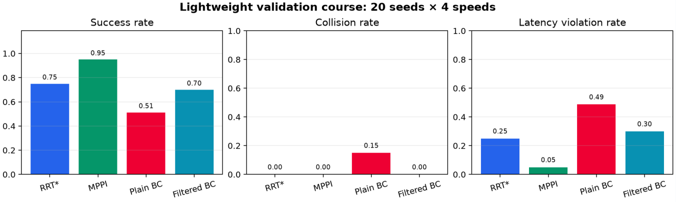
- 2,205 samples.
- Positive rate: 0.3383.

Small MLP vs TTC

- PR-AUC: 0.885 vs 0.784.
- Risk recall: 0.992.
- False negative rate: 0.008.

Insight. For rare safety events, PR-AUC, recall and false negatives are more meaningful than accuracy.

RQ4: validation harness ranks candidates before heavy simulation

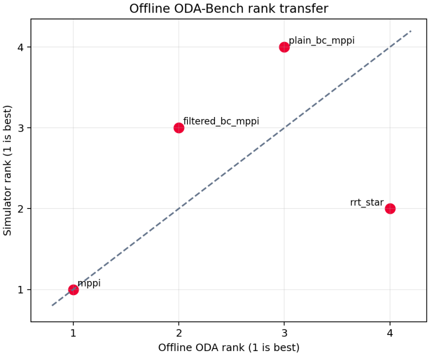


Current run

- 20 seeds $\times$ 4 speed levels.
- Backend: kinematic fallback.
- MPPI has highest success: 0.95.
- Plain BC shows worst collision: 0.15.

The gym-pybullet-drones backend is implemented with CtrlAviary + DSLPIDControl, but PyBullet did not build on the current Apple Silicon environment. The GTX machine should run the true PyBullet validation.

RQ5: offline ranking currently agrees only partially with validation



Rank transfer

- Spearman $\rho = 0.4$ on current fallback validation.
- Offline rank 1 and validation rank 1 both select MPPI.
- BC methods diverge more, exposing failure cases that need simulator checks.

Insight. ODA-Bench is useful as an early filter; PyBullet validation is still needed before stronger claims.

Discussion: what ODA-Bench can and cannot claim today

What the evidence supports

- ODA can be converted into repeatable offline safety labels.
- Planner, BC and risk tracks run on a 300-trial trial-level split.
- Filtered BC reduces violation.
- Small MLP risk predictor improves PR-AUC over TTC.

What remains limited

- Offline geometry does not replace closed-loop dynamics.
- Kinematic fallback is not equivalent to PyBullet or real UAV flight.
- ODA scenes are limited by available obstacle metadata and sensor noise.
- Offline RL is a future track, not a main result yet.

Conclusion: ODA-Bench is a practical bridge before field testing

Core message

- We propose an evaluation protocol and benchmark suite built on top of ODA.
- It supports offline planner evaluation, lightweight policy training and risk prediction.
- It reduces dependence on simulation-heavy workflows for early candidate screening.

Next step

- Run the implemented gym-pybullet-drones backend on the GTX 1050 Ti machine.
- Replace fallback validation with PyBullet ranks.
- Add TD3+BC/IQL only if the simulator validation is stable.

●● **Insight.** The final paper claim should be benchmark + evaluator + tasks + baselines + lightweight validation, not field-ready UAV autonomy.